

ADITYA KUMAR ARYA

adityaarya803@gmail.com | +91 8789337007 | [LinkedIn](#) | [GitHub](#) | 30/08/2004

PROFILE SUMMARY

Data Science and Machine Learning enthusiast with experience building end-to-end ML pipelines, predictive systems, and deployed AI applications. Proficient in Python, Scikit-learn, TensorFlow, FastAPI, and cloud deployment, with hands-on exposure to research internships and production-grade ML solutions.

EDUCATION

Sant Longowal Institute of Engineering and Technology

Sangrur, Punjab

BE in Electronics and Communication Engineering | *GPA: 9.26*

Nov 2022 – Jun 2026 (Expected)

EXPERIENCE

Project Trainee | Bhabha Atomic Research Centre (BARC)

Jun 2025 – Aug 2025

- Developed **Physics-Informed Neural Network (PINN)** models in MATLAB for damped oscillators and nuclear reactor point kinetics by embedding governing differential equations via automatic differentiation, achieving accurate simulations with up to 90% less data and strong alignment with analytical benchmarks.

Project Intern | AICTE – TechSaksham (Microsoft & SAP)

Jan 2025 – Feb 2025

- Engineered a robust prediction system for Diabetes, Heart Disease, and Parkinson's utilizing Advanced Machine Learning Algorithms, achieving **91% accuracy** on 20,000+ records, and deploying the solution as a Streamlit web app under the Microsoft & SAP TechSaksham program (executed by Edunet Foundation).

PROJECTS

AI Learning Intelligence Tool *FastAPI, CLI, Scikit-learn, Pandas, Render*

[Source Code](#)

- Architected a production-grade **ML pipeline** for ingestion, feature engineering, training, and inference on 1,000+ learner interaction records, ensuring reproducibility through robust logging and exception handling.
- Operationalized an executable AI system with CLI and FastAPI REST interfaces, enabling real-time and batch inference for 100+ learners per run and explainable course completion and early dropout predictions.

Restaurant Rating Prediction *ML Pipelines, Flask, Matplotlib, Seaborn, Plotly, Render*

[Source Code](#)

- Implemented a modular ML pipeline on 30,000+ restaurant reviews with integrated logging, utility, and exception handling, improving reproducibility across 5+ runs and 20% fewer preprocessing errors.
- Built and deployed a digital solution on Render; performed **A/B testing** and iterative parameter tuning (heuristics-based) to optimize performance, achieving **92%** deployment accuracy.

Retail Sales Forecasting System *XGBoost, Prophet, Flask, Pytest*

[Source Code](#)

- Built an end-to-end time-series forecasting pipeline on ~840K retail records with lag and rolling features, leveraging XGBoost as a baseline and hybrid XGBoost-Prophet approach to achieve an RMSE of ~900.
- Deployed a Flask-based web application on Render, enabling real-time predictions using 16 engineered features within a modular pipeline, ensuring reliable and maintainable inference.

SKILLS

Programming:	C/C++, Python, Java, SQL
Development:	TensorFlow, PyTorch, NumPy, Pandas, Scikit-learn, OpenCV, NLP, LLM
Data Science & Machine Learning:	EDA, Feature Engineering, Model Training, Evaluation & Optimization
Statistics & Mathematics:	Descriptive Statistics, Probability & Distributions, Hypothesis Testing
Cloud & Deployment:	AWS, Docker, Render, Streamlit, Microsoft Azure, Rest API Development
Tools & Technologies:	MySQL, Power BI, VSCode, Jupyter Notebook, Git/GitHub
Soft Skills:	Cross-Functional Collaboration, Technical Mentorship & Communication

ACHIEVEMENTS & CERTIFICATIONS

- Pursuing a Minor Degree in **Computer Science** and **AI/ML**.
- Tata Imagination Challenge (TIC) 2023:** Global Top 2400 & Semifinalist.
- Swaraj Mahindra Scholarship Program:** Scholar, 2022.
- Solved 300+ problems on different coding platforms.
- 5 stars SQL proficiency with SQL (Advanced) Certification on HackerRank.

[CodeChef](#)

[HackerRank](#)

POSITION OF RESPONSIBILITY

Student Placement Representative

- Spearheading campus placements for 450+ students and streamlining recruitment processes.